

# Jia Li | Curriculum Vitae

[Jia.Li@utdallas.edu](mailto:Jia.Li@utdallas.edu) • [jiali-home.github.io](https://jiali-home.github.io) • [Google Scholar](#)

## Education

---

- **The University of Texas at Dallas** *Aug. 2023 – Expected May 2027*  
Ph.D. in Computer Science, Ph.D. Candidate. Advised by [Yapeng Tian](#).
- **Sichuan University** *Sep. 2018 – Jun. 2022*  
B.S. in Mathematics.

## Research Interests

---

**Research Areas:**    **Computer Vision**    •    **Artificial Intelligence**    •    **Machine Learning**

I develop AI assistants for the physical world that can understand the user, interpret the surrounding physical and social environment, and operate reliably over time. My research focuses on egocentric and multimodal perception, causal and streaming reasoning, and human-centered assistance.

## Research Experience

---

- **The University of Texas at Dallas – Graduate Research Assistant** *Aug. 2024 – Present*  
Conduct research on multimodal and egocentric AI, with current work in streaming procedural understanding, online error detection, and intent-conditioned affordance grounding.
- **AT&T – Research Scientist Intern** *May 2026 – Aug. 2026*  
Developed a physics-guided AI framework for Passive Intermodulation (PIM) detection and diagnosis in cellular networks. Combined spectrum-based ML, physics-driven candidate generation, and agentic reasoning to identify likely intermodulation orders and carrier combinations. Built a synthetic benchmark and research prototype, improving Top-1 diagnosis accuracy from 20.9% to 65.2%, and presented the work internally at AT&T.

## Teaching Experience & Development

---

- **The University of Texas at Dallas – Instructor of Record** *Aug. 2026 – Dec. 2026*  
*ECS 1100: Introduction to Engineering and Computer Science* (~30 students). Collaboratively designed course content, in-class activities, assignments, and assessment rubrics as part of the instructional team, and independently delivered weekly instruction for my assigned section. Topics included responsible AI, professional communication, career preparation, and engineering decision-making; responsibilities also included grading, office hours, and student communication.
- **The University of Texas at Dallas – Teaching Assistant** *Aug. 2023 – Dec. 2024*  
*Operating Systems & Discrete Mathematics* Held office hours, managed Piazza-based student support, and graded assignments, exams, and programming projects. Mentored students on course projects and coordinated student project presentations.
- **The University of Texas at Dallas – Graduate Teaching Certificate** *2025*  
Completed evidence-based teaching coursework and workshops, a faculty observation of my teaching, and pedagogical work: full course syllabi for *Discrete Mathematics* and *Data Structures and Algorithms Laboratory*, designed with backward design and learning outcomes aligned to Bloom's taxonomy.
- **The University of Texas at Dallas – Learning to Teach in Engineering** *Jan. 2025 – May 2025*  
Graduate course on evidence-based STEM teaching, built around Felder & Brent, *Teaching and Learning STEM: A Practical Guide*. Observed a large-enrollment Discrete Mathematics lecture and a Digital Systems laboratory, and interviewed each instructor about their course design.

## Publications

---

### Peer-Reviewed Publications

- **How Much Future Helps? A Controlled Study of Future-Privileged Supervision for Causal Egocentric Gaze Estimation**  
CVPR 2026 Workshops — GAZE (Best Paper Award) [\[link\]](#)  
[Jia Li](#), Wenjie Zhao, Fnu Atisri, Sanskriti Aripineni, Shijian Deng, Jon E. Froehlich, Yuhang Zhao, Yapeng Tian
- **Do Audio-Visual Segmentation Models Truly Segment Sounding Objects?**  
AAAI 2026 [\[link\]](#)  
[Jia Li](#), Wenjie Zhao, Ziru Huang, Yunhui Guo, Yapeng Tian
- **Talking to Me or Someone Else? Rethinking Talk-to-Me Detection in Egocentric Videos**  
ACM Multimedia 2026 [\[link\]](#)  
Feiyu Du, Xi He, [Jia Li](#), Yapeng Tian, Weili Wu
- **A Skill-Augmented Agentic Framework and Benchmark for Multi-Video Understanding**  
EMNLP 2026 — Findings [\[link\]](#)  
Yue Zhang, Liqiang Jing, [Jia Li](#), Yapeng Tian, Xinya Du, Yunhui Guo, Vibhav Gogate
- **Mitigating the ID-OOD Tradeoff in Open-Set Test-Time Adaptation**  
CVPR 2026 — Findings [\[link\]](#)  
Wenjie Zhao, [Jia Li](#), Xin Dong, Yapeng Tian, Yu Xiang, Yunhui Guo
- **Toward Gaze Target Detection of Young Autistic Children**  
AAAI 2026 — AI for Social Impact Track [\[link\]](#)  
Shijian Deng, Erin E. Kosloski, Siva Sai Nagender Vasireddy, [Jia Li](#), Randi Sierra Sherwood, Feroz Mohamed Hatha, Siddhi Patel, Pamela R. Rollins, Yapeng Tian
- **What to Distinguish and How? Opportunities and Challenges of Augmenting Multiple, Cluttered Objects in Complex Scenes for People with Low Vision**  
ASSETS 2026 [\[link\]](#)  
Yuheng Wu, Ruijia Chen, Jaewook Lee, [Jia Li](#), Kexin Zhang, Meng Fong Lio, Weibing Wang, Sanbrita Mondal, Jon E. Froehlich, Yapeng Tian, Yuhang Zhao
- **CURE-OOD: Benchmarking Out-of-Distribution Detection for Survival Prediction**  
TMLR 2026 [\[link\]](#)  
Wenjie Zhao, [Jia Li](#), Mingrui Liu, Jing Wang, Yunhui Guo
- **Segment Every Out-of-Distribution Object**  
CVPR 2024 [\[link\]](#)  
Wenjie Zhao, [Jia Li](#), Xin Dong, Yu Xiang, Yunhui Guo
- **AV-Mamba: Cross-Modality Selective State Space Models for Audio-Visual Question Answering**  
CVPR 2024 Workshops — Sight and Sound [\[link\]](#)  
Ziru Huang, [Jia Li](#), Wenjie Zhao, Yunhui Guo, Yapeng Tian
- **Measuring Aleatoric and Epistemic Uncertainty in LLMs: Empirical Evaluation on ID and OOD QA Tasks**  
KDD 2024 Workshops — UDM [\[link\]](#)  
Kevin Wang\*, Subre Abdoul Moktar\*, [Jia Li\\*](#), Kangshuo Li\*, Feng Chen

### Preprints and Surveys

- **ARGaze: Autoregressive Transformers for Online Egocentric Gaze Estimation**  
Under review at TMLR, 2026 [\[link\]](#)  
[Jia Li](#), Wenjie Zhao, Shijian Deng, Bolin Lai, Yuheng Wu, Ruijia Chen, Jon E. Froehlich, Yuhang Zhao, Yapeng Tian
- **From Waveforms to Pixels: A Survey on Audio-Visual Segmentation**  
Preprint, 2025 [\[link\]](#)  
[Jia Li](#), Yapeng Tian

- **From Pixel to Mask: A Survey of Out-of-Distribution Segmentation**  
Preprint, 2025 [\[link\]](#)  
Wenjie Zhao, [Jia Li](#), Yunhui Guo

## Honors, Awards & Research Support

---

- **Gold Reviewer Award, ICML,** 2026
- **Best Paper Award, CVPR GAZE Workshop** 2026
- **Google Cloud Research Credits, \$1,000** 2025
- **Lambda Research Grant Sponsorship, \$1,000** 2025
- **Outstanding Reviewer Award, NeurIPS,** 2024

## Professional Service

---

- **Conference Program Committee / Reviewer:**
  - IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)
  - IEEE/CVF International Conference on Computer Vision (ICCV)
  - Conference on Neural Information Processing Systems (NeurIPS)
  - International Conference on Learning Representations (ICLR)
  - International Conference on Machine Learning (ICML)
  - AAAI Conference on Artificial Intelligence (AAAI)
  - International Conference on Artificial Intelligence and Statistics (AISTATS)
  - ACM International Conference on Multimedia (ACM Multimedia)
  - ACM CHI Conference on Human Factors in Computing Systems (CHI)
  - ACL Rolling Review (ARR)
- **Journal Reviewer:**
  - Transactions on Machine Learning Research (TMLR)
  - IEEE Transactions on Multimedia

## Selected Talks, Presentations & Panels

---

- **How Much Future Helps? A Controlled Study of Future-Privileged Supervision for Causal Egocentric Gaze Estimation**  
*Oral presentation*, CVPR GAZE Workshop, 2026.
- **ARGaze: Autoregressive Transformers for Online Egocentric Gaze Estimation**  
*Invited research talk*, Prof. Yi Ding's group, The University of Texas at Dallas, 2026.
- **Graduate Teaching and Research Assistant Experience Panel**  
*Invited panelist*, The University of Texas at Dallas, 2026.
- **Panel Discussion on Career Paths after Graduation**  
*Invited panelist*, IEEE, 2026.

## Student Mentoring

---

- **Abdullah Khajamohideen – Undergraduate Researcher** May 2026 – Aug. 2026  
Mentored research on *EgoAffordance: Egocentric Part Affordance for Everyday Object Interaction*. Guided dataset development, semantic part and affordance annotation, benchmark design, and preliminary model evaluation. The project resulted in a research poster through the Undergraduate Research Apprenticeship Program (URAP).

- **Fnu Atisri – Undergraduate Researcher** *May 2025 – Aug. 2025*  
Mentored research on ECOGaze, a controlled study of future-privileged supervision for causal egocentric gaze estimation, which received the Best Paper Award at the CVPR 2026 GAZE Workshop.
- **Sanskriti Aripineni – Undergraduate Researcher** *May 2025 – Aug. 2025*  
Co-mentored research on *Real-Time Egocentric Gaze Estimation*, serving as the primary day-to-day research mentor. Guided causal video models for online gaze prediction. The project resulted in a research poster through the Undergraduate Research Apprenticeship Program (URAP) and contributed to ECOGaze (CVPR 2026 GAZE Workshop Best Paper Award).
- **Kevin Wang & Subre Abdoul Moktar – Undergraduate Researchers** *Jan. 2024 – Aug. 2024*  
Co-mentored a two-student team on aleatoric and epistemic uncertainty in large language models, from experimental design through paper development. The work was accepted to the 3rd UDM Workshop at ACM SIGKDD 2024.